

8 Individuals develop artificial immunity either actively or passively.

(a) The table gives some statements about immunity.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show whether these statements are true for these types of immunity.

(2)

Statement	Type of artificial immunity			
	both active and passive	active only	passive only	neither active nor passive
Antigens are injected into the person	☐	☐	☐	☐
Immunity is long term	☐	☐	☐	☐

(b) Vaccination against a virus results in the production of antibodies.

Describe how antibodies are produced following activation of T cells.

(2)

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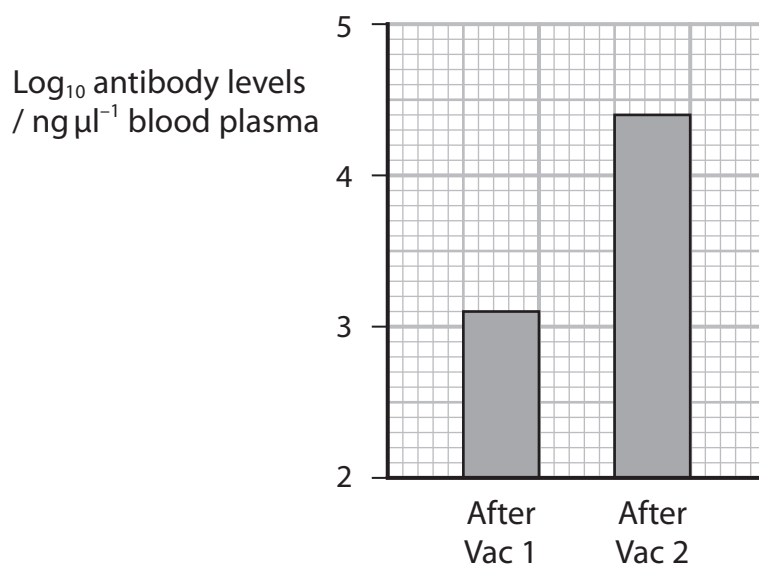
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(c) The levels of antibodies following vaccinations were investigated.

There were two groups of people. Group 1 **had not** been infected with the virus, before the vaccine was given. Group 2 **had** been infected with the virus before the vaccine was given.

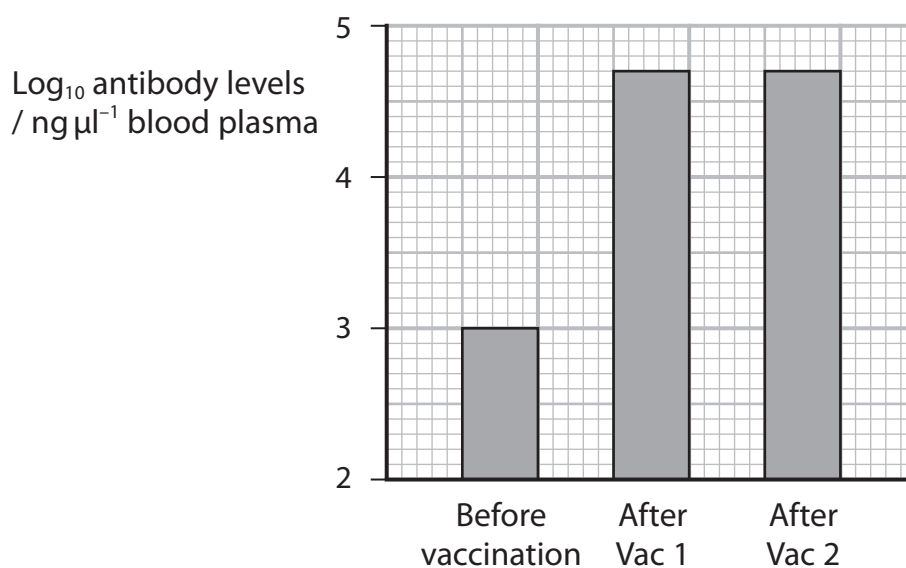
Both groups of people received two vaccinations, Vac 1 and Vac 2.

The graphs show the results of this investigation.



Group 1:

People who had not been infected before the vaccine was given



Group 2:

People who had been infected with the virus before the vaccine was given

- (i) Explain **two** ways in which this investigation would have to be controlled to provide valid data.

(2)

- (ii) In group 1, Vac 2 increased the levels of antibody more than Vac 1.

Calculate how many times greater this increase was.

(2)

Answer

(iii) Suggest why there is no data on the graph for group 1 before vaccination.

(1)

*(iv) Explain the results of this investigation.

(6)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 90 MARKS